Paperboard cartons: HMTc Standard

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CORRESPONDENCE

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PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for paperboard cartons in Food-Contact Consumer Goods and Kitchenware, evaluated on specific element migration INTO FOOD OR A FOOD SIMULANT from the polymeric food-contact surface, µg/kg of food, determined by the Reg (EU) 10/2011 Annex III to V simulant, temperature and contact-time conventions. This is a MIGRATION basis and not a total-content basis: no cell on this row limits how much of an element the material CONTAINS, and no cell on this row is expressed per kg of material.. 7 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable sovereign maximum for this exact product form expressed as specific element migration into food or a food simulant; a disclosed read-across from the nearest applicable sovereign value in the same basis where the food-contact material family is expressly excluded from or not reached by that instrument; a substantiated not-material determination; or a control. No cell is occurrence-derived, and no cell borrows a value across a basis.

2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	10	human-grade read-across (European Union)

Cadmium (Cd)	Tier 1	2	human-grade read-across (European Union)
Total arsenic (tAs)	Tier 2	10	human-grade read-across (European Union)
Inorganic arsenic (iAs)	Tier 1	n/m	not material in this matrix
Mercury, total (tHg)	Tier 1	10	human-grade read-across (European Union)
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	20	human-grade read-across (European Union)
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	1,000	human-grade read-across (European Union)
Total chromium (Cr)	Tier 2	10	human-grade read-across (European Union)
Hexavalent chromium (Cr-VI)	Tier 2	n/m	not material in this matrix

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	10	European Union — human-food analog
Cadmium (Cd)	2	European Union — human-food analog
Total arsenic (tAs)	10	European Union — human-food analog

Inorganic arsenic (iAs)	n/m	Inorganic arsenic is a dietary speciation with no separate food-contact-material instrument anywhere in the credible set: every food-contact element requirement is written on total arsenic. Arsenic migrating from a polymeric food-contact article is in any case inorganic, so the total-arsenic non-detect requirement is simultaneously the tightest achievable inorganic-arsenic ceiling and bounds this cell by identity. No separate inorganic-arsenic ceiling is published, and inorganic arsenic never substitutes for the total-arsenic cell.
Mercury, total (tHg)	10	European Union — human-food analog

Methylmercury (MeHg)	n/m	Methylmercury is formed by microbial methylation in aquatic food chains, where an anaerobic sulfate- or iron-reducing microbial community acts on a bioavailable inorganic-mercury pool over residence times measured in weeks to years. The determination on this row is argued from THIS ROW'S OWN MATERIALS rather than from a generic cured-polymer clause. The food-contact surface of a coated carton is a cured polyolefin layer with no organic carbon available to a methylating community. The row's substrate and its uncoated configuration are cellulosic board, an organic wettable matrix, so the argument is made about the board and not only about the polymer: methylation requires a persistent anaerobic niche, a sulfate-reducing or iron-reducing community and a bioavailable inorganic-mercury pool, and a dry-stored, single-use or aseptically filled carton supplies none of the three. Its contact is short and aerobic, its service life is orders of magnitude shorter than methylation kinetics, and cellulose, the wet-strength and sizing chemistry of liquid-packaging board and recovered-fibre furnish supply no methyl donor to mercury. No credible sovereign sets a methylmercury requirement for a food-contact material of any family in any basis. The total-mercury requirement on this row is a published non-detect value rather than a control, so any organomercury fraction is bounded by a real number, and the board component is separately tested for recovered-fibre element load under the component plan. Methylmercury is not material and never substitutes for the total-mercury cell.
Nickel (Ni)	20	European Union — human-food analog

Tin (Sn)	control	Regulation (EU) 10/2011 Annex II lists no tin entry, and no credible sovereign sets a tin migration or release ceiling for a food-contact article of any material family. Tin is nonetheless a mandatory panel cell, because tin stabilisers in PVC compounds and tin-bearing solder are live pathways in parts of this category, so it is tested on every certified lot with no invented ceiling. This cell carries NO companion: the only sovereign tin numbers in existence are maximum levels on canned food, which are a FOOD basis on the contents of a tinplate container, and they are neither applicable to this row nor publishable as a migration ceiling. They are named as a companion only on the rows where a tinplate or crown component actually exists.
Aluminium (Al)	1,000	European Union — human-food analog
Total chromium (Cr)	10	European Union — human-food analog
Hexavalent chromium (Cr-VI)	n/m	Regulation (EU) 10/2011 Annex II point 2 controls hexavalent chromium through the total-chromium requirement rather than through a separate species limit: total chromium must be not detectable, and the relaxed 3.6 mg/kg food value becomes available only where the operator documents that chromium(VI) is absent. HMTc declines that relaxation, so every certified article is held to total chromium not detectable at 0.01 mg/kg food, which bounds hexavalent chromium at or below the same value. No credible sovereign sets a hexavalent-chromium migration limit for a food-contact material, so no separate species ceiling is published; hexavalent chromium never substitutes for the total-chromium cell.

4 | Cite this standard

Pendergrass, K. "Paperboard cartons." In Heavy Metal Tested & Certified (HMTc) Food-Contact Consumer Goods and Kitchenware standard, version 1.0 (2026-08-17). Institute of Contaminant Standards (ICS), 2026.

References

[1] European Union — human-food maximum levels
Binds Pb, Cd, tAs, tHg, Ni, Al, Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/food--packaging-paperboard-cartons. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-food-packaging-paperboard-cartons.